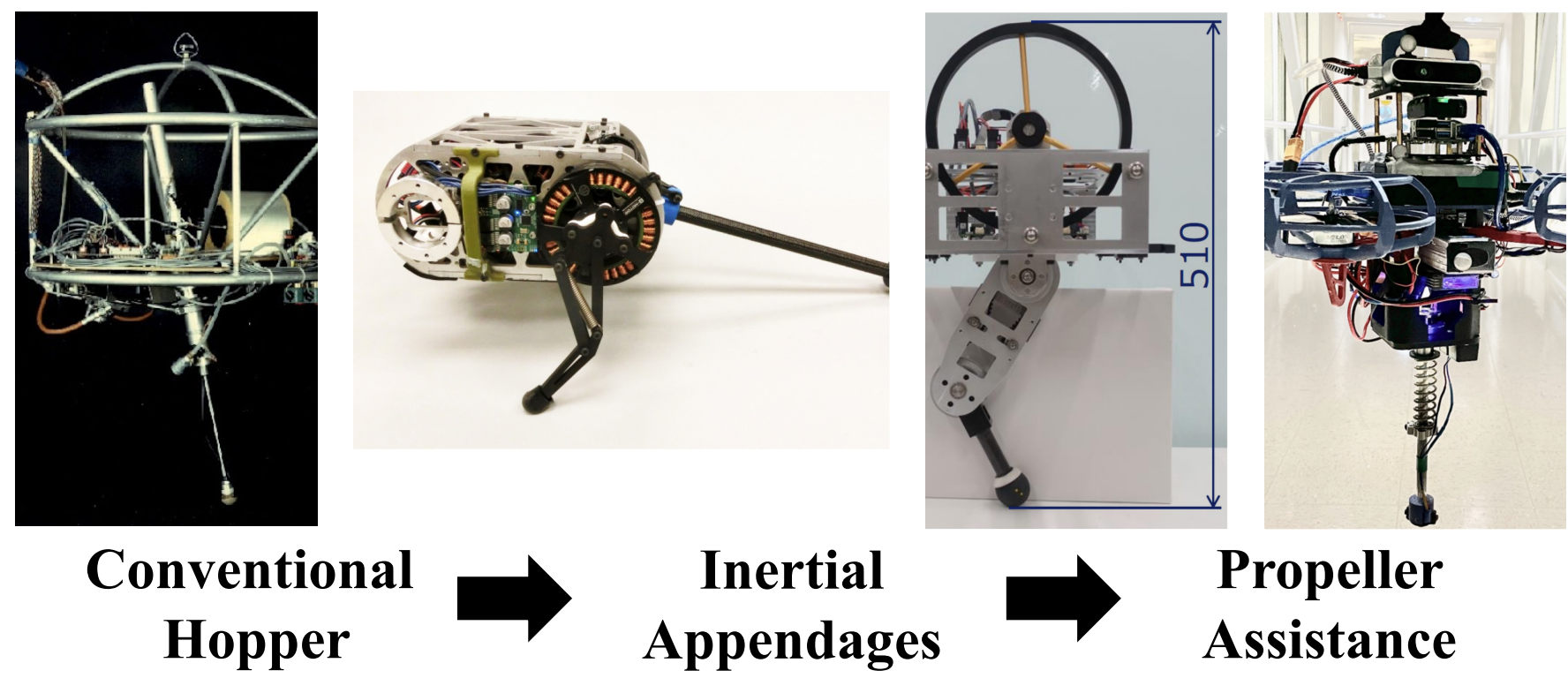


# Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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PAPER + ARXIV2026 IEEE 32nd International Conference on Automation  
Science and Engineering (CASE)17-21 AUGUST 2026  
SHENYANG, CHINA

## 1 Problem- hybrid dynamics create an authority gap



### ■ Stance

Ground reaction force exists only during a short contact interval.

### ■ Flight

No contact: attitude becomes underactuated.

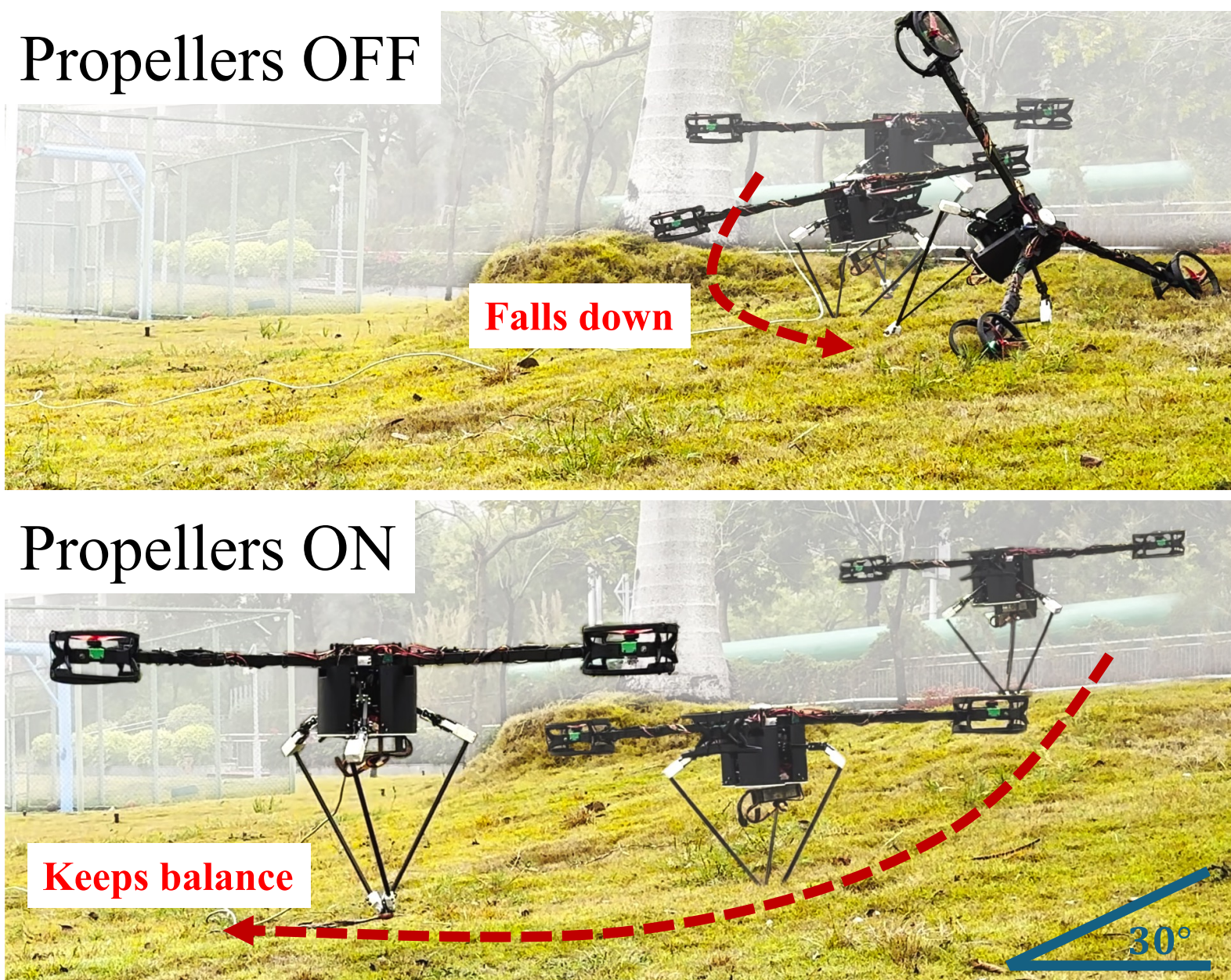
### ■ Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

## 2 First proof- stable on 30° grass



### ■ Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

### ■ Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

### ■ What this establishes

Residual authority in stance.

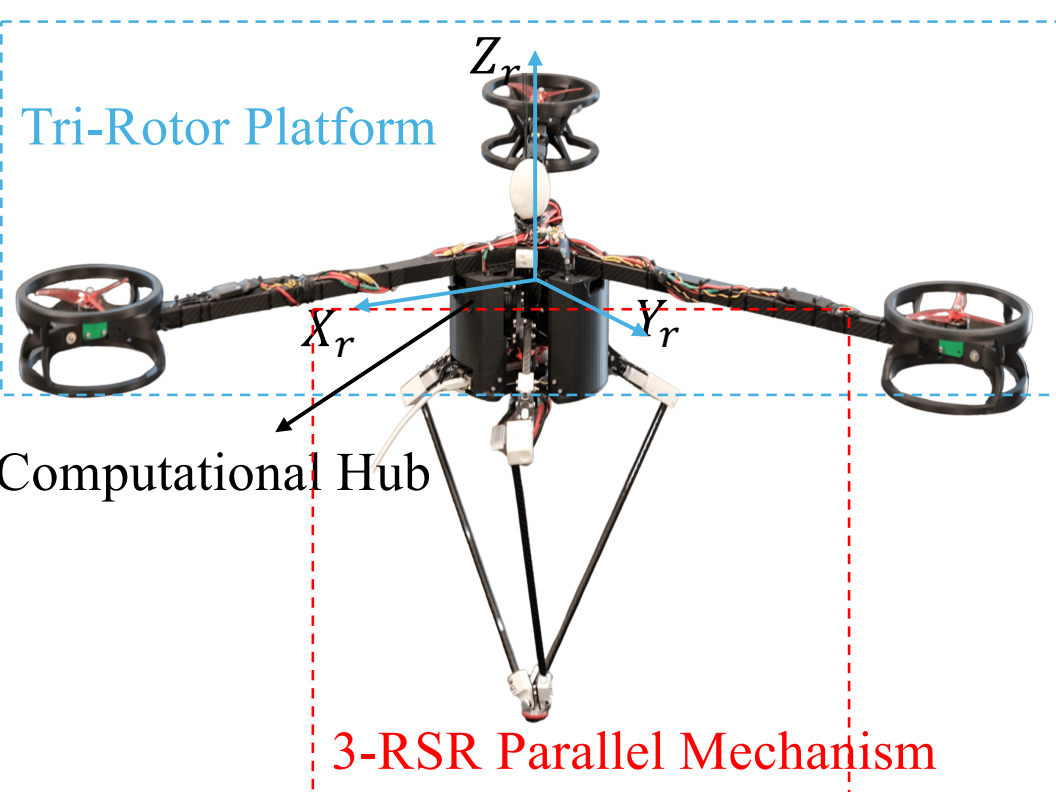
Primary attitude authority in flight.

Faster recovery from uneven impacts.

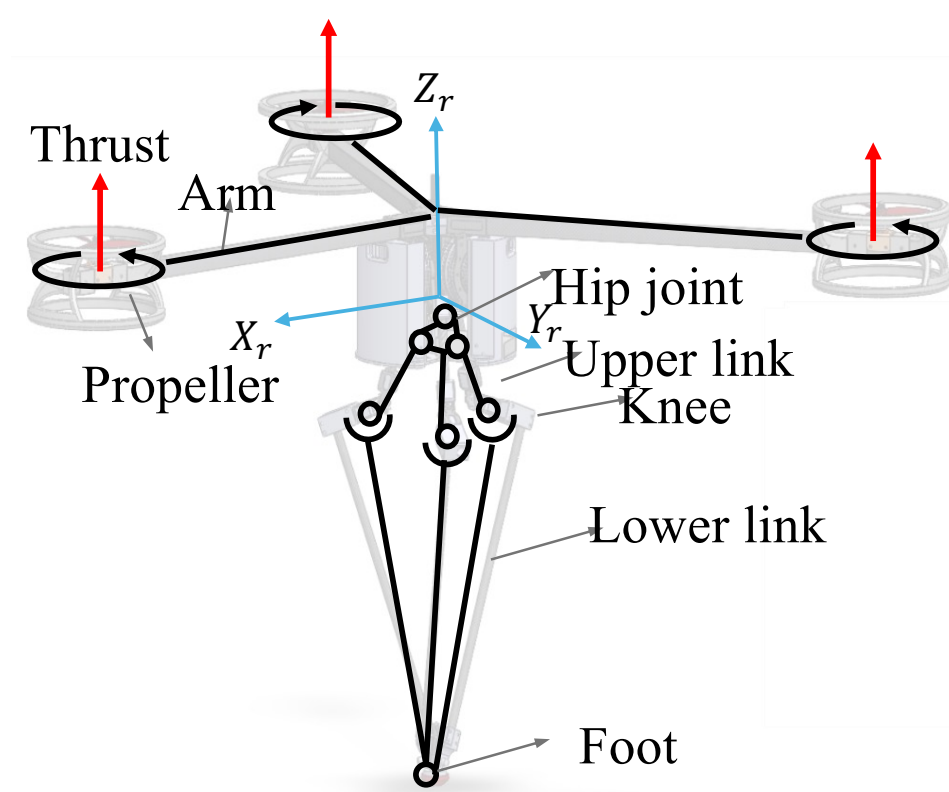
Not primary lift: the rotors close the attitude-authority gap.

Propellers off: attitude diverges and the robot falls. Propellers on: it remains upright and keeps hopping.

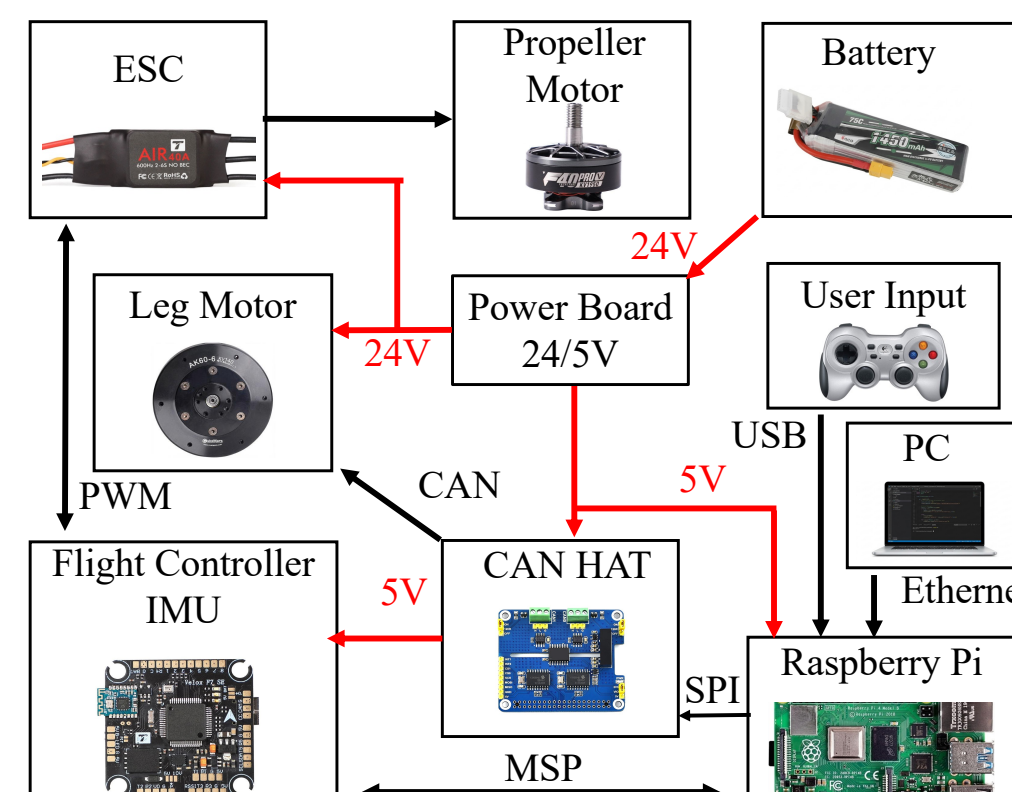
## 3 Platform- hardware built for leg-first hopping



Robot design



Leg and tri-rotor components



Electronic architecture

### ■ 3-RSR parallel leg

Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

### ■ Computational hub

Batteries and electronics near the hip: low center of mass, low inertia.

### ■ Y-layout tri-rotor

Continuous attitude authority; **10 N** per rotor, total 0.82 mg: a hopper, not a drone.

### ■ External PC @ 500 Hz

State machine, estimation, wrench planning, and the HFA allocator.

### ■ Raspberry Pi interface

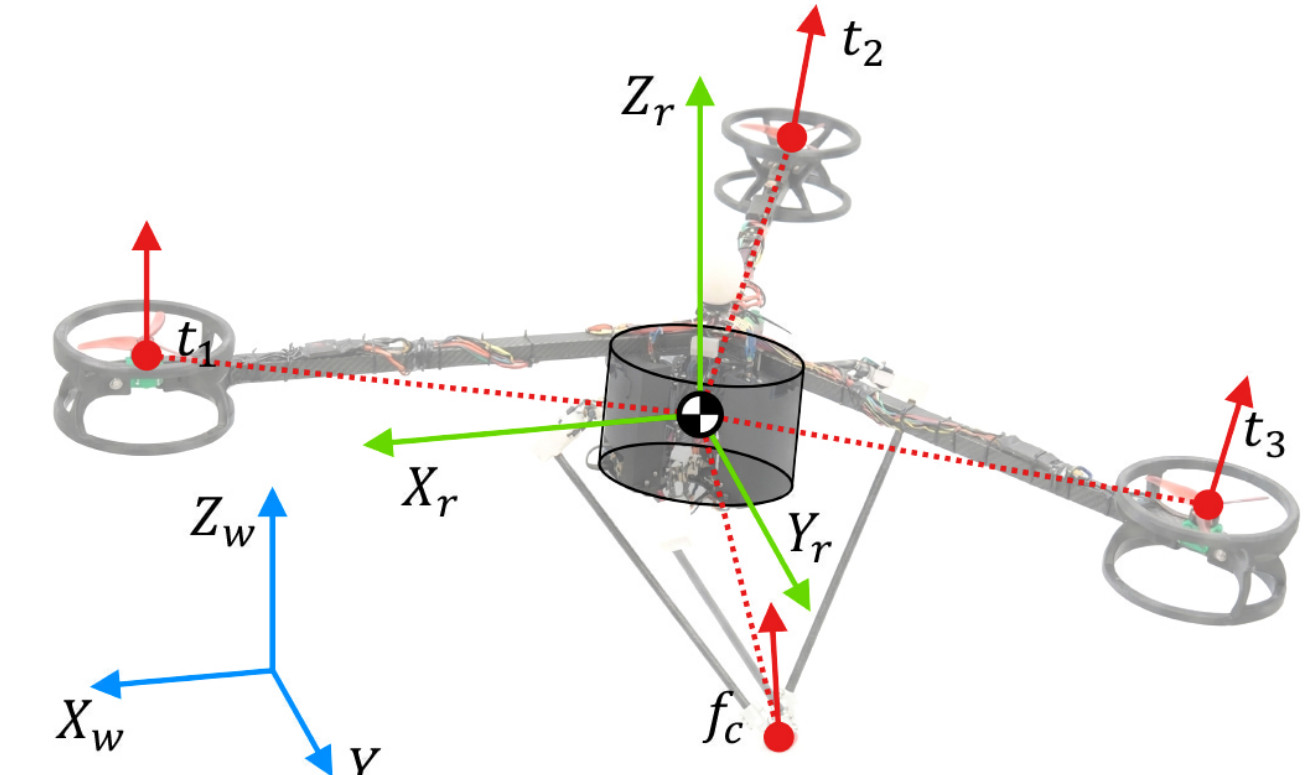
CAN to the three leg motors; MSP to the flight controller.

### ■ Real-time link

LCM over Ethernet: RTT  $\approx 0.2$  ms, 99%  $< 0.5$  ms, within the 2 ms period.

3.75 kg total ■ 9.2% leg mass ■ 20 Nm / leg motor ■ 500 Hz control

## 4 Model- one rigid body, two actuator sets

Contact force  $f_c$  and thrusts  $t_{1,2,3}$  produce the body wrench.

### ■ Net wrench = leg contact + three rotors

$$\begin{bmatrix} F_{net} \\ \tau_{net} \end{bmatrix} = \begin{bmatrix} f_c \\ r_c \times f_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} d_i \\ r_i \times d_i \end{bmatrix} t_i$$

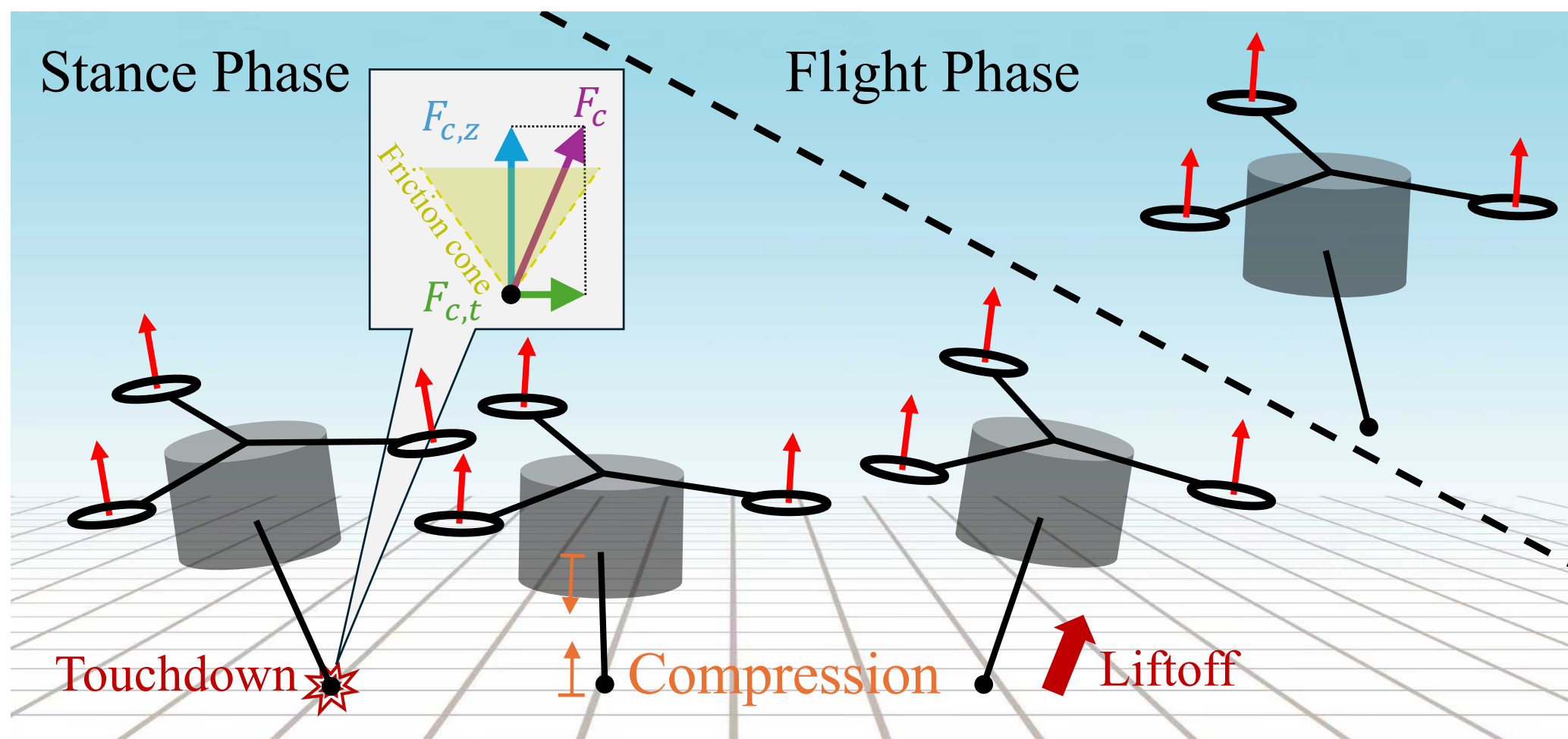
### ■ Newton-Euler dynamics of the CoM

$$ma = F_{net} + mg, \quad \dot{\omega} = I^{-1}(R^T \tau_{net} - \omega \times I \omega)$$

### ■ Symbols (values used in all experiments)

 $f_c$ : ground reaction force (world) $r_c, r_i$ : arms: CoM  $\rightarrow$  contact, rotor  $i$  $t_i \in [0, 10]$  N thrust along  $d_i = R e_3$  $F_{net}, \tau_{net}$ : net force, moment $m=3.75$  kg;  $g$  gravity;  $I$  inertia $R \in \text{SO}(3)$  attitude;  $\omega$  body rate $J$  leg Jacobian;  $\tau_{max}=20$  Nm limit $\mu=0.4$  friction coefficient $p_z, \dot{p}_z$ : CoM height, velocity;  $h_{des}=0.664$  m $l$  leg length;  $l_0=0.464$  m;  $\Delta_{td}$  threshold $k_z, b_z, k_E$  vertical,  $k_v, k_r$  Raibert gains $K_R, K_\omega$  attitude PD gains

## 5 Onehop- event-driven authority handoff



### ■ Touchdown: $l \leq l_0 - \Delta_{td}$

Leg absorbs impact; energy law drives the next apex.

### ■ Liftoff: $l > l_0$

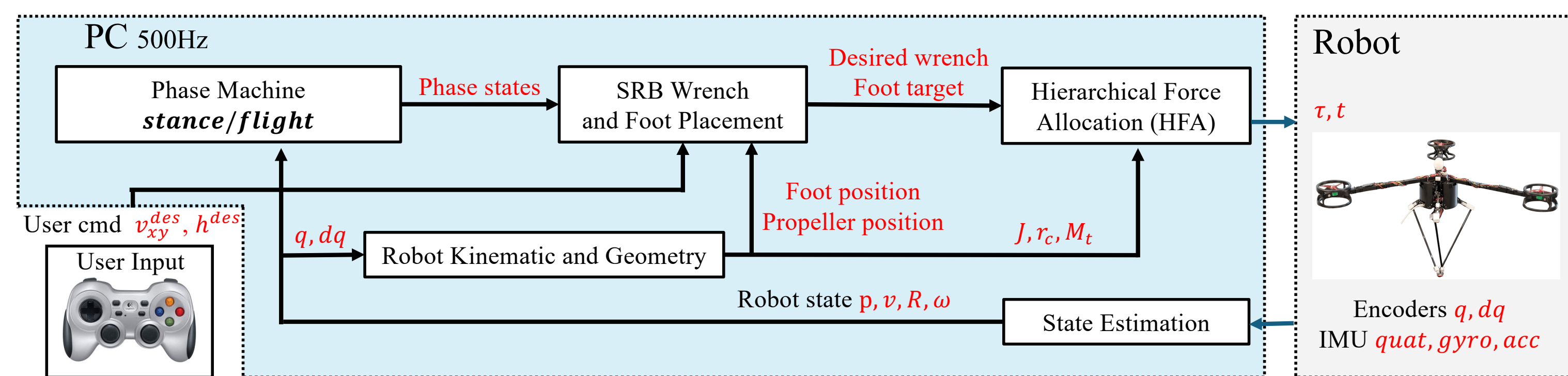
Contact wrench vanishes; rotors primary for attitude.

### ■ Handoff

Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

## 6 Control- leg-priority hierarchical force allocation

500 Hz architecture: commands + feedback  $\rightarrow$  planner  $\rightarrow$  two-stage allocator  $\rightarrow$  leg torques + rotor thrusts.

### ■ A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + mg p_z, \quad E_{des} = mg h_{des}$$

$$f_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

### ■ B. SO(3) attitude PD (both phases)

$$e_R = \frac{1}{2} (R_{des}^T R - R^T R_{des})^v$$

$$\tau_b^{des} = -K_R e_R + K_\omega (\omega^{des} - \omega)$$

$$\text{Roll/pitch only; yaw left free. } \tau^{des} = R \tau_b^{des}.$$

### ■ C. Leg first - constrained least squares

$$f_{c,xy}^{cmd} = \arg \min_{f_{xy}} \|(r_c \times f_c)_{rp} - (\tau^{des})_{rp}\|_2^2$$

$$\text{s.t. } \|f_{xy}\|_2 \leq \mu f_{c,z}^{des}, \quad \|J^T(-R^T f_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

### ■ D. Rotors take only the residual

$$\tau_{res}^{cmd} = \tau^{des} - r_c \times f_c^{cmd}$$

$$\delta \tau^{cmd} = \arg \min_{\delta \tau} \|(M_t \delta \tau)_{rp} - (\tau_{res}^{cmd})_{rp}\|_2^2$$

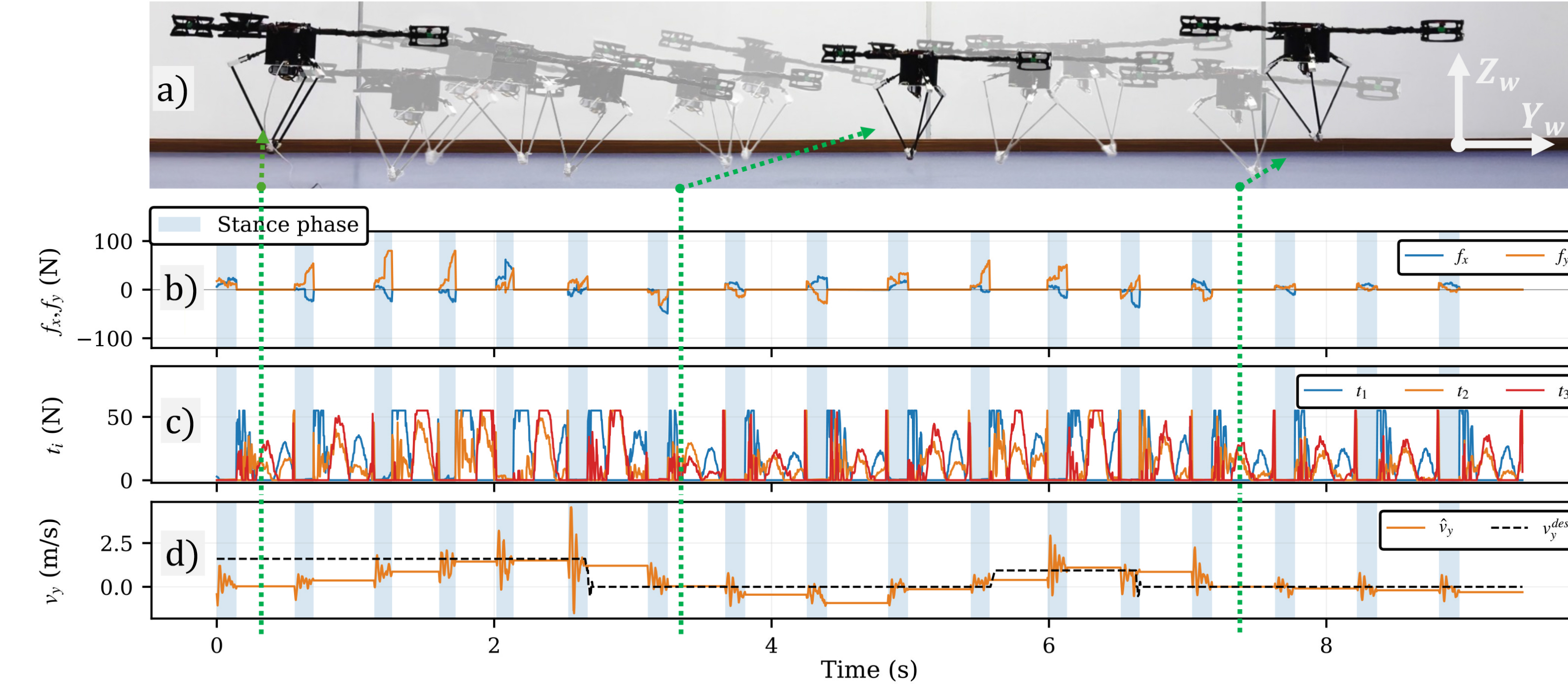
$$\text{s.t. thrust bounds; } t^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta \tau^{cmd}, T_{base} = 0.02 \text{ mg.}$$

### ■ E. Flight - rotors primary for attitude; Raibert-style foot placement

$$f_c = 0; \quad \text{rotor track } (\tau^{des})_{rp}; \quad r_{td,xy}^{des} = k_v v_{xy} - k_r v_{xy}^{des} \Rightarrow p_{foot}^{des} = [x_{td}^{des}, y_{td}^{des}, -\sqrt{l_0^2 - \|r_{td,xy}^{des}\|_2^2}]^T$$

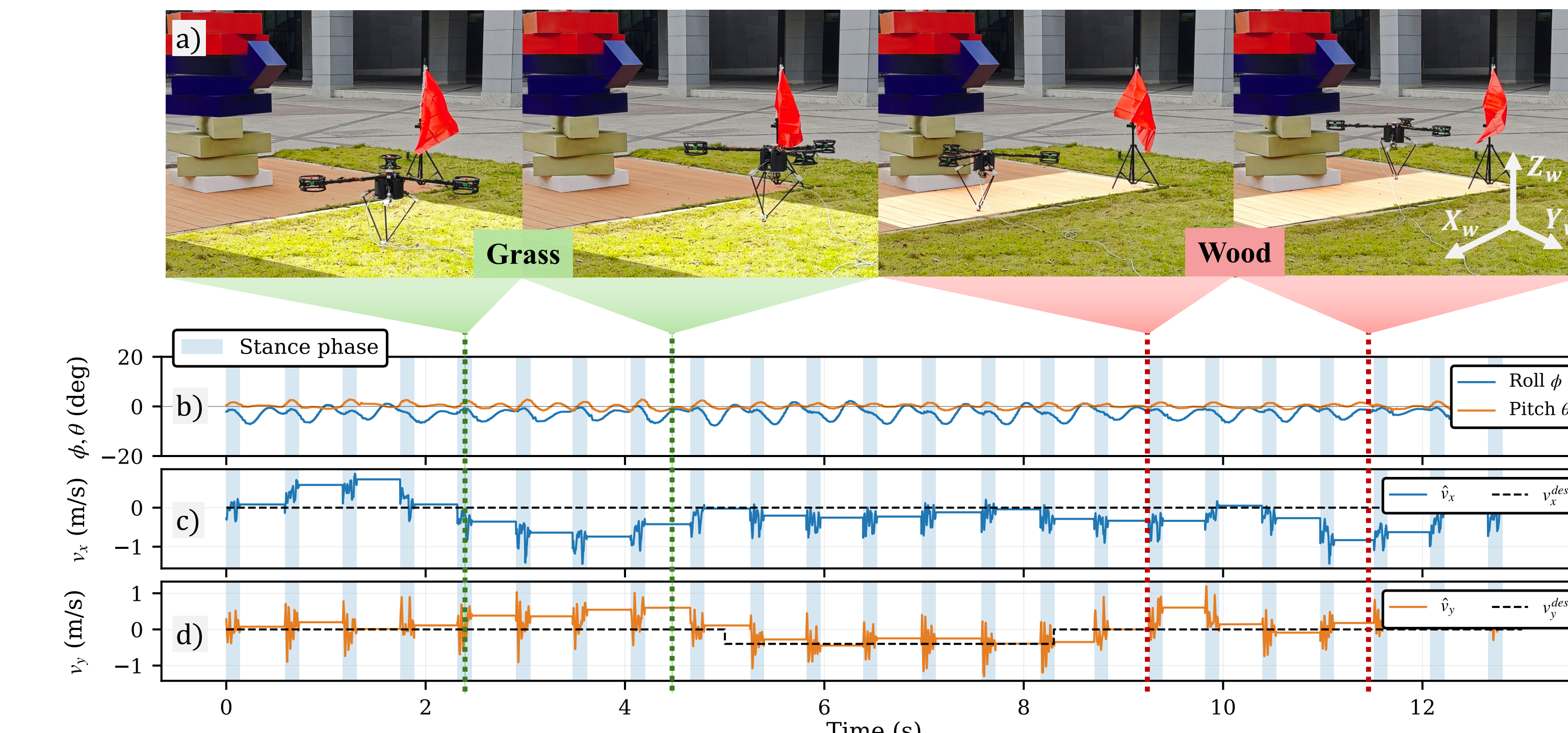
By construction the rotors can only add to - never replace - the leg's contact action.

## 7 Validation- indoor stop-and-go tracking

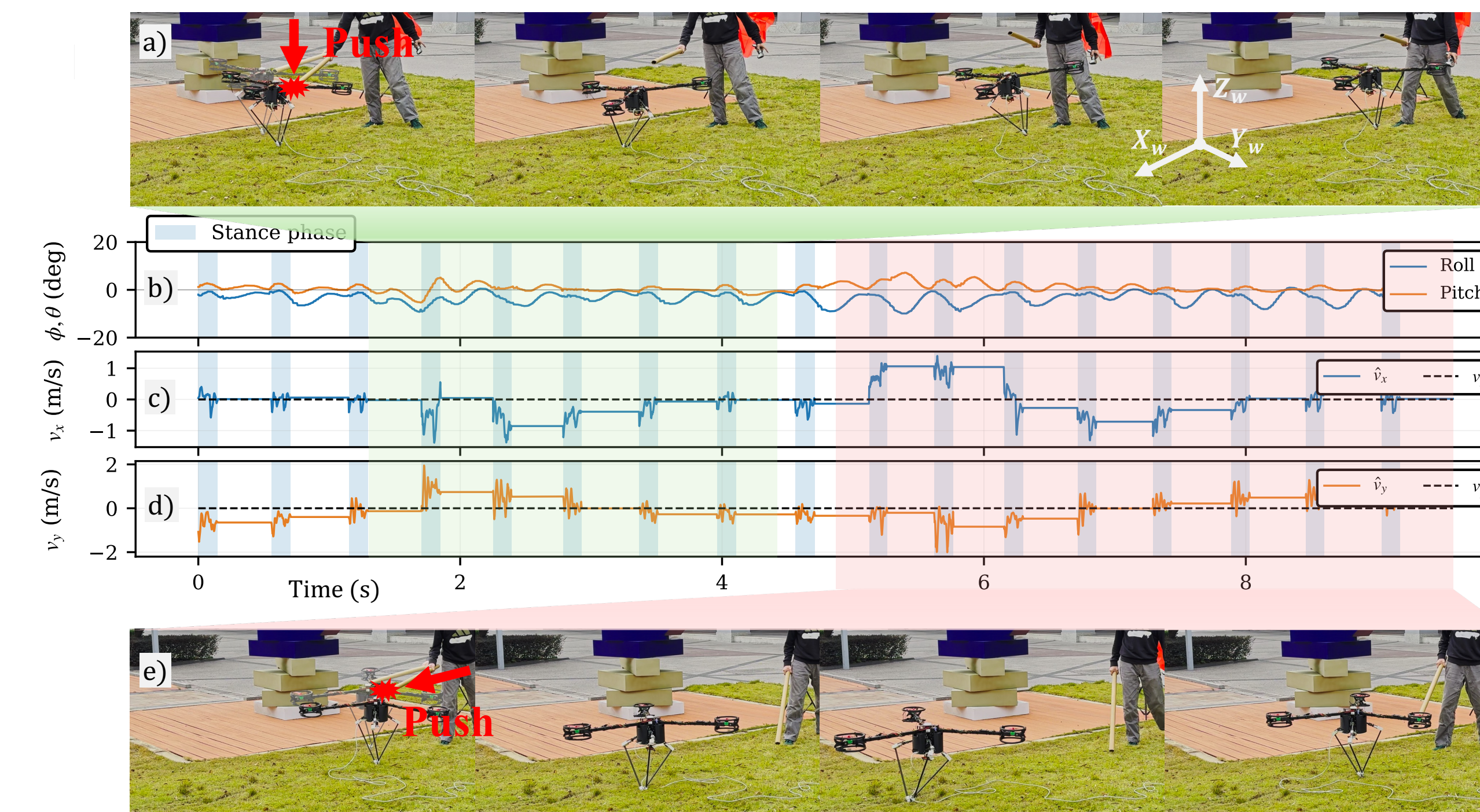


Stop-and-go commands: accelerate, then hard stop (peaks 1.6 and 0.94 m/s) - 17 hops. Velocity recovered stride to stride by Raibert-style foot placement (Panel 6, E).

## 8 Validation- grass to wood

Grass  $\rightarrow$  wood under light wind: **23 hops**- same controller, no terrain model, no gain retuning. Attitude stays tight: roll  $\sigma = 2.1^\circ$ , pitch  $\sigma = 1.0^\circ$ ; deviations decay within 1-2 hops.

## 9 Validation- push recovery

Rotor-arm push: propellers restore roll/pitch in flight - stable again in **1-2 hops**. Body push ( $\approx 1.4$  m/s velocity spike): foot placement recovers tracking in **3 hops**.